

Backward Design - Course Foundations Planner

Course: Systems Engineering in Public Programs

Stage 1 - Desired Results

Learning Outcomes

Design mathematical problems by using appropriate assumptions and strategies based on different stakeholders' needs.
Evaluate whether model solutions prioritize feasibility or optimality.
Select modeling tools based on problem requirements and justify tool selection
Adapt decision-model strategies to different contexts.
Evaluate how modelling choices might have unintended consequences.

Transfer

What kinds of long-term independent accomplishments are desired?
Learners will be able to independently use their learning to...
Use the **modeling strategies** tools to design effective decision-making models, extracting reasonable assumptions and stating things ignored.
Recognize the implications of **modeling choices** to outcomes of decisions in social settings.
Design adequate **questions** to the problem based on **stakeholders** and **expected outcomes**.
Identify how the **language** changes and affects different contexts in interdisciplinary research.
Find at least one appropriate tool to use to answer at least one of the questions.
Regularly reflect on and evaluate the **effects of their assumptions on the final modelling results**.

Meaning

UNDERSTANDINGS

What, specifically, do you want learners to understand? What inferences should they be able to make?
Learners will understand that...
All representations of complex problems involve simplifying assumptions that both enable analysis and limit applicability
Social interactions and diverse, sometimes competing, goals in social problems contexts add complexity to technical solutions.
Not all feasible solutions are optimal solutions, and not all optimal solutions can be effectively implemented in practice
Each question can only be answered with specific tools and modeling strategies
A good model answers a specific but not broad question

ESSENTIAL QUESTIONS

What thought-provoking questions will foster inquiry, meaning-making, and transfer?
Learners will consider ...
What is a good model? What does this mean?
What is the problem we want to solve? Who is going to benefit from it?
What is the difference in challenges of modeling social and technical problems?
What are the trade-offs one makes while transforming real problem to a mathematical one?
How well can we effectively answer our questions?
How realistic is the answer the model is giving?
Is a good model for one, a good model for all problems?

Acquisition

What facts and basic concepts should learners know and be able to recall?
Learners will know...
Stages of mathematical modeling design
Procedures to inquire modelling outcomes
Types of modeling strategies
Types of solving tools

What discrete skills and processes should learners be able to use?
Learners will be skilled at...
Translation of practical problems to mathematical ones.
Selecting appropriate solving tools
Selecting an appropriate perspective to answer questions.

Stage 2 - Evidence		
Evaluative Criteria		Assessment Evidence
THOROUGHNESS Are all desired results being assessed? All Transfer Goals All Meaning Goals All Acquisition Goals	EVALUATIVE CRITERIA What criteria will be used in each assessment to evaluate the attainment of the desired results? Regardless of the format of the assessment, what qualities are most important? For detailed information about criteria, see assignment descriptions. Criteria includes, but is not limited to: - Completeness - Alignment (e.g., instructional components) - Critical assessment - Clarity on presentation and report	PERFORMANCE TASK(S): How will learners demonstrate their understanding (meaning-making and transfer) through complex performance? <i>Learners will show that they really understand by evidence of ...</i> One mathematical model that follows requirements. A complete report with graphs on expected results for selected modeling choices by midterm. A clear presentation detailing assumptions, stakeholders and initial decisions by midterm. A complete report detailing decisions made, and results. A clear presentation and results. Peer-discussion with instructor-led inquiries to show clarity on problem requirements and instructions every class. Justification of each choice based on stakeholders and assumptions made. OTHER EVIDENCE: What other evidence will you collect to demonstrate whether Stage 1 Results were achieved? <i>Learners will also show they achieved Stage 1 results by...</i> Participation in class, by answering inquiries and participating in group discussions.